MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

B.Tech. COURSE PLAN

Theory Course

Department:	**BIOTECHNOLOGY**			
Course Name & code:	**BIOINFORMATICS & BIO 3151**		**Core**	
Semester & branch:	**V SEM**	**B.Tech Biotechnology**		
Name of the faculty:	**Dr S Balaji**			
No of contact hours/week:	**L**	**T**	**P**	**C**
	3	**1**	**0**	**4**

COURSE OUTCOMES (COs)

	At the end of this course, the student should be able to:	**No. of Contact Hours**	**Marks**
CO1	Solve computational problems related to biological data	16	30
CO2	Analyze and interpret molecular data	12	30
CO3	Integrate bioinformatics with other disciplines (especially research)	10	30
CO4	Apply practical aspects for sequence/structure analysis	10	10
	Total	**48**	**100**

ASSESSMENT PLAN

Components	In-Semester Assessment Components (ISAC)			End semester/Makeup examination
Duration				180 minutes
Weightage	50%			50%
Typology of questions	Applying; Analysing;	Applying; Analysing;	Applying; Analysing. Evaluating. Creating	Applying; Analysing; Evaluating; Creating
Pattern	**MISAC 1** **MISAC 2** **MISAC 3** **FISAC**	05 marks 30 marks 05 marks 10 marks	To decide by the Faculty. May be Surprise assignments, Quiz, Take home assignments, Seminar, Mini project, Assignment etc.	Answer all 5 full questions of 10 marks each. Each question may have 2 to 3 parts of 3/4/5/6/7 marks
Schedule			As per academic schedule	
Topics covered	**MISAC 1** **MISAC 2** **MISAC 3** **FISAC**	Aug 28-Sep 2 Sep 25-30 Oct 9-14 Oct30-Nov06	As per academic schedule	Comprehensive examination covering the full syllabus. Students are expected to answer all questions

LESSON PLAN

L No	TOPICS	Course Outcome Addressed
L0	Teaching methodology & course introduction	0
L1	Bioinformatics overview	CO2
L2	The Digital code of Life, information theory and biology	CO1
L3	The evolutionary basis of sequence alignment	CO2
L4	Detecting ORFs, protein Vs nucletotide sequence alignment	CO1
L5	Homology, orthology and paralogy	CO2
L6	Conservation, conserved substitution and semi-conserved substitution	CO2
L7	Dot plots, scoring matrices	CO1
L8	Gaps and Gap penalties	CO1
L9	Substitution Matrices: PAM and BLOSUM	CO1
L10	Introduction to biological sequence databases	CO2
L11	The Gene Bank database, Flat file, Format and Content	CO1
L12	Database similarity searching: FASTA & BLAST	CO1
L13	Practical aspects of fine-tuning parameters in local alignment	CO1
L14	Introduction to computational biology	CO1
L15	Dynamic programming algorithm	CO1
L16	Problem Solving: Global and Local alignments	CO2
L17	Statistical and Biological significance of Alignments	CO2
L18	Multiple Sequence Alignment	CO2
L19	PCR Primer Design	CO2
L20	Primer design practical considerations	CO1
L21	Practical aspect of Sequence Alignment	CO2
L22	Sequence assembly: Overlap-Layout-Consensus approach	CO2
L23	Sequence assembly – issues and challenges	CO2
L24	Sequence repeats and complexity in sequence assembly	CO1
L25	Problem Solving: Sequence assembly	CO1
L26	Graph theory & networks	CO1
L27	chemical graphs and amino acid comparisons	CO1
L28	Walk-path-trail-circuit-cycle; Eulerian and Hamiltonian paths	CO1
L29	Detecting Functional Sites in the DNA (Gene finding)	CO3
L30	Evaluation of gene finding tools	CO4
L31	Secondary structure, HP-lattice model	CO3
L32	PDB file format	CO3
L33	Motifs and domains; symmetry of proteins	CO3
L34	Structure of nucleic acids	CO3
L35	Molecular visualization tools	CO4
L36	Protein structure-function relationships	CO3
L37	Protein modeling	CO3
L38	Homology modeling: practical aspects	CO3
L39	Structure validation: Ramachandran plot and other methods	CO3
L40	Homology Modeling: Applications and Practical aspects	CO4
L41	Evolution	CO3
L42	Synapomorphy, plesiomorphy and symplesiomorphy	CO4
L43	Phylogenetics on the web, some simple practical considerations	CO4
L44	Morphology based cladogram	CO4

L45	Building Phylogenetic Trees (Basics and Steps involved)	CO4
L46	Phylogenetic software and Tree evaluation	CO4
L47	Phylogenetic network	CO4
L48	Bioinformatics: future aspects	CO4

References:

- Andreas D Baxevanis. 2004. BIOINFORMATICS – A practical Guide to the Analysis of Genes and Proteins. Wiley Interscience.
- David R. Westhead. 2003. Instant Notes: Bioinformatics. BIOS Scientific Publishers Ltd.
- David W Mount. 2001. BIOINFORMATICS: Sequence and Genome Analysis. Cold Spring Harbor
- Arthur M. Lesk. 2002. Introduction to Bioinformatics. Oxford University Press

COURSE LEARNING OUTCOMES (CLOs)

CO	At the end of this course, the student should be able to:	No. of Contact Hours	Marks	Program Outcomes (PO's)	Program Specific Outcomes (PSO's)	** Learning Outcomes (LOs)	Blooms Taxonomy (BL)
CO1	Solve computational problems related to biological data	16	30	PO1-5, PO8-10, PO12	1		4
CO2	Analyze and interpret molecular data	12	30	PO1-5, PO8-10, PO12	1		3
CO3	Integrate bioinformatics with other disciplines (especially research)	10	30	PO1-5, PO8-12	1		3
CO4	Apply practical aspects for sequence/structure analysis	10	10	PO1-5, PO8-10, PO12	1		3, 4
	Total		**100**				

Course Articulation Matrix

CO/CLO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	1	2	1	3	1	1	1	1	2	1		2	1	1
CO2	2	1	1	1	1	1		1	1	1		1	1	1
CO3	1	2	1	1	1	1		2	1	1		1	1	1
CO4	1	1	1	1	1			1	1	1		1	1	1
Average Articulation Level	1	2	1	1	1	1	1	1	1	1	NA	1	1	1

Submitted by:

(Signature of the faculty)

Date:

Approved by:

(Signature of HOD)

Date:

FACULTY MEMBERS TEACHING THE COURSE (IF MULTIPLE SECTIONS EXIST):

FACULTY	SECTION	FACULTY	SECTION